MAT 3007 – Optimization

The Simplex Method and Initialization

Lecture 09

Instructor: Junchi Yang

CUHK-Shenzhen

Recap

Simplex Method: Neighborng BFS

Neighbors and Principle Idea:

- Two basic solutions are neighbors if they differ in exactly one basic index.
- Basic idea: Move from one BFS to a neighboring BFS while improving the objective function.

How to Move to a Neighbor:

- Move along j-th basic directions from a BFS $\boldsymbol{x}$:

$$\boldsymbol{x} \to \boldsymbol{x} + \theta \boldsymbol{d}, \quad \boldsymbol{d} = [\boldsymbol{d}_B; \boldsymbol{d}_N] = [-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0].$$

- The definition of basic direction guarantees feasibility $\boldsymbol{A}(\boldsymbol{x} + \theta \boldsymbol{d}) = \boldsymbol{b}$ for all θ. Therefore, we need to choose θ to ensure $\boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}$.

Reduced Costs & Selecting Directions: $c^\top(x+\theta d) - c^\top x = \theta c^\top d$

- The reduced cost for any j is defined as

Note $\bar{c}_j = 0$ if $j \in B$

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j = \boldsymbol{c}^\top \boldsymbol{d}.$$

$-d_B$

- Moving from $\boldsymbol{x}$ to $\boldsymbol{y} = \boldsymbol{x} + \theta \boldsymbol{d}$, where $\boldsymbol{d}$ is j-th basic direction, reduces the objective by $\theta \bar{c}_j$.

Theorem: Stopping Criterion

Let $\boldsymbol{x}$ be a BFS associated with the basis $B(1), \ldots, B(m)$ and let $\bar{\boldsymbol{c}}$ be the corresponding vector of reduced costs. If $\bar{\boldsymbol{c}} \geq 0$, then $\boldsymbol{x}$ must be optimal.

- It means that if we can not find a neighboring BFS that improves the objective function, then the current BFS is an optimal solution.

Choose a Stepsize: Minimum-Ratio-Test

Specifying θ:

- Select j-th basic direction $\boldsymbol{d}$ with $\bar{c}_j < 0$. Move along $\boldsymbol{d}$ as much as possible:

$$\theta^* = \max\{\theta \geq 0 : \boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}\}.$$

Choose a Stepsize: Minimum-Ratio-Test

Specifying θ:

- Select j-th basic direction $\boldsymbol{d}$ with $\bar{c}_j < 0$. Move along $\boldsymbol{d}$ as much as possible:

$$\theta^* = \max\{\theta \geq 0 : \boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}\}.$$

We consider two cases:

$x \to x + \theta^* d$, function value is reduced by $\theta^* \bar{c}_j \to \infty$ as $\theta^* \to \infty$.

1. If $\boldsymbol{d} \geq \boldsymbol{0}$, then $\theta^* = \infty$. The original LP is unbounded.

Choose a Stepsize: Minimum-Ratio-Test

Specifying θ:

- Select j-th basic direction $\boldsymbol{d}$ with $\bar{c}_j < 0$. Move along $\boldsymbol{d}$ as much as possible:

$$\theta^* = \max\{\theta \geq 0 : \boldsymbol{x} + \theta \boldsymbol{d} \geq \boldsymbol{0}\}.$$

We consider two cases:

1. If $\boldsymbol{d} \geq \boldsymbol{0}$, then $\theta^* = \infty$. The original LP is unbounded.
2. If $d_i < 0$ for some i, then we can solve:

$$\theta^* = \min_{\{i : d_i < 0\}} -\frac{x_i}{d_i} = \min_{\{i \in B : d_i < 0\}} -\frac{x_i}{d_i}.$$

Moving to a New Basis

Assume θ^* is finite (otherwise the problem is unbounded), then we can move to another feasible point:

$$\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}.$$

Let $B(\ell) \in B$ be an index with $\theta^* = -x_{B(\ell)}/d_{B(\ell)}$.

$$y_{B(\ell)} = x_{B(\ell)} + \theta^* d_{B(\ell)} = 0.$$

Moving to a New Basis

Assume θ^* is finite (otherwise the problem is unbounded), then we can move to another feasible point:

$$\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}.$$

Let $B(\ell) \in B$ be an index with $\theta^* = -x_{B(\ell)}/d_{B(\ell)}$.

$$y_{B(\ell)} = x_{B(\ell)} + \theta^* d_{B(\ell)} = 0.$$

Thus, the basic variable $x_{B(\ell)}$ has become zero, whereas the non-basic variable x_j is now positive (equal to θ^*). The basis has changed to: (It remains to prove the correpsonding columns of A linearly independent.)

$$B(1), \ldots, B(\ell-1), j, B(\ell+1), \ldots, B(m).$$

Theorem: Properties of y

Let $\boldsymbol{x}$ be a nondegenerate BFS ($\boldsymbol{x}_B > 0$) with basic indices B and let $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$ be generated by the simplex iteration. Then, $\boldsymbol{y}$ is a basic feasible solution associated with the basic indices

$$\bar{B} = \{B(1), \ldots, B(\ell-1), j, B(\ell+1), \ldots, B(m)\}.$$

proof idea: consider the following matrix

$$A_B^{-1} A_{\bar{B}} = \begin{pmatrix} | & | & & | & & | \\ A_B^{-1}A_{B(1)} & A_B^{-1}A_{B(2)} & \cdots & A_B^{-1}A_j & \cdots & A_B^{-1}A_{B(m)} \\ | & | & & | & & | \end{pmatrix}$$

$A_B^{-1} A_{B(i)} = e_i$ because $A_{B(i)} = A_B e_i$

$$= \begin{pmatrix} | & | & & | & & | \\ e_1 & e_2 & \cdots & -d_B & \cdots & e_m \\ | & | & & | & & | \end{pmatrix}$$

Observation: ① ℓ-th coordinate of $e_i = 0$ for $i \neq \ell$

② ℓ-th coordinate of d_B, i.e. $d_{B(\ell)}$ is negative since $\underbrace{x_{B(\ell)}}_{>0} + \theta^* d_{B(\ell)} = 0$.

$\Rightarrow$ $A_{\bar{B}}^{\rightarrow} A_{\bar{B}}$ have linearly indep. columns.

$\Rightarrow$ $A_{\bar{B}}$ have linearly indep. columns. $\square$

An Iteration of the Simplex Method

Initialization: We start from a BFS $\boldsymbol{x}$ (with basic set B).

1. Compute the reduced costs $\bar{\boldsymbol{c}}$:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$

   - If none of the reduced costs is negative, then $\boldsymbol{x}$ is optimal.
   - Otherwise choose some j such that $\bar{c}_j < 0$.

An Iteration of the Simplex Method

Initialization: We start from a BFS $\boldsymbol{x}$ (with basic set B).

1. Compute the reduced costs $\bar{\boldsymbol{c}}$:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$

   - If none of the reduced costs is negative, then $\boldsymbol{x}$ is optimal.
   - Otherwise choose some j such that $\bar{c}_j < 0$.

2. Compute the jth basic direction

$$\boldsymbol{d} = [-\boldsymbol{A}_B^{-1} \boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0]$$

   - If $\boldsymbol{d} \geq \boldsymbol{0}$, then the problem is unbounded
   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0} \{-\frac{x_i}{d_i}\}$.

An Iteration of the Simplex Method

Initialization: We start from a BFS $\boldsymbol{x}$ (with basic set B).

1. Compute the reduced costs $\bar{\boldsymbol{c}}$:
$$\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_j.$$
   - If none of the reduced costs is negative, then $\boldsymbol{x}$ is optimal. ← algorithm stops
   - Otherwise choose some j such that $\bar{c}_j < 0$.
2. Compute the jth basic direction
$$\boldsymbol{d} = [-\boldsymbol{A}_B^{-1}\boldsymbol{A}_j; 0; \ldots; 1; \ldots; 0]$$
   - If $\boldsymbol{d} \geq \boldsymbol{0}$, then the problem is unbounded ⇐ algorithm stops
   - Otherwise, compute $\theta^* = \min_{i \in B, d_i < 0} \{-\frac{x_i}{d_i}\}$.
3. Set $\boldsymbol{y} = \boldsymbol{x} + \theta^* \boldsymbol{d}$. Then, $\boldsymbol{y}$ is the new BFS with index j replacing $B(\ell)$ in the basis, where $B(\ell)$ is the index attaining the minimum in θ^*. The objective function value is changed by $\theta^* \boldsymbol{c}^\top \boldsymbol{d} = \theta^* \bar{c}_j$.
4. Repeat these procedures.

Theorem: Convergence

Assume that the feasible set is nonempty and that every BFS is non-degenerate. Then, the simplex method terminates after a finite number of iterations with the following options:

1. The method stops with an BFS which is an optimal solution.
2. We have found a vector $\boldsymbol{d}$ with $\boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}$, $\boldsymbol{d} \geq \boldsymbol{0}$, and $\boldsymbol{c}^\top \boldsymbol{d} < 0$, and the optimal value is $-\infty$.

Degeneracy

Degeneracy

We call a basic feasible solution $\boldsymbol{x}$ (non-)degenerate if (none) some of the basic variables are 0.

$$\theta^* = \min_{\{i \in B \mid d_i < 0\}} -\frac{x_i}{d_i}.$$

if $x_i = 0$ and $d_i \geq 0$
$\Rightarrow x_i + \theta d_i \geq 0 \rightarrow \theta$ can still be positive.

If for some $i \in B$ with $x_i = 0$ and $d_i < 0$, then we obtain $\theta^* = 0$. Hence $\boldsymbol{y} = \boldsymbol{x}$ and the objective value reamins unchanged.

Change of basis: We still change the basic index from $B(\ell)$ ($B(\ell)$ leaving the basis) to j (j entering the basis) and proceed to the next iteration.

- There might be a cycle, i.e., visit the same BFS more than once. (Cycle only happens with degeneracy. Why?) We need a proper pivoting rule to avoid it.

Pivoting Rules: Choosing the Entering Basis

In the algorithm, we can choose an index $j \in N$ with negative reduced cost to enter the basis. Sometimes, there are more than one j with $\bar{c}_j < 0$. In this case, we need to introduce some rules to choose the entering basis.

Here are several possible rules:

1. **Smallest Index Rule**: Choose the smallest index j with $\bar{c}_j < 0$.
2. **Most Negative Rule**: Choose the smallest $\bar{c}_j$.
3. **Most Decrement Rule**: Choose j with the smallest $\theta^* \bar{c}_j$.

Pivoting Rules: Choosing the Exiting Basis

We choose an index $i \in B$ that attains the minimum

$$\theta^* = \min_{\{i \in B | d_i < 0\}} -\frac{x_i}{d_i}.$$

to leave the basis. It is possible that there is more than one index that attains the minimum (tie).Then we also need a rule to decide the exiting basis.

- The most commonly used rule is the smallest index rule.

When this tie happens, the next BFS will be degenerate. (Why?)

Bland's Rule

Theorem: Bland's Rule

If we use the smallest index rule for choosing both the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.

- Proof in Bertsimas + Tsitsiklis.
- Using the Bland's rule when applying the simplex method, we can guarantee to stop within a finite number of iterations at an optimal solution.

Stopping Rule: Revisited

Theorem: Reduced Costs and Uniqueness

Let $\boldsymbol{x}$ be a BFS with associated (non-)basic indices B (and N).

- If the reduced costs of every non-basic variable is positive, i.e., $\bar{\boldsymbol{c}}_N > 0$, then $\boldsymbol{x}$ is a unique optimal solution.
- If $\boldsymbol{x}$ is a unique optimal solution and non-degenerate, then the reduced cost of every non-basic variable is positive.

- The second result is no longer true when $\boldsymbol{x}$ is degenerate (can you find counter examples?)

Finding an Initial BFS

Finding an Initial BFS

Denote $\tilde{A} = [A \ I_m]$

$Ax \le b$, $x \ge 0$ → standard form: $Ax + s = b$, $x \ge 0, s \ge 0$ → $Ax + I \cdot s = b$, $x \ge 0, s \ge 0$.

Assume $b \ge 0$.

So far, we assumed that we start with a certain BFS: $\begin{pmatrix} x \\ s \end{pmatrix} = \begin{pmatrix} 0 \\ b \end{pmatrix}$ is a BFS because:

① $A \cdot 0 + I \cdot s = b$

② $x = 0$, $b \ge 0$

- This can be done easily if the standard form is derived by adding slacks to each constraint. (Why?)

However, in general, it is not necessarily easy to get an initial BFS from the standard form. For example:

③ consider b as basic variable. The corresponding columns $\tilde{A}$ are I_m, which are linearly indep.

$$
\begin{array}{lrrrrcl}
\text{minimize} & x_1 & +x_2 & +x_3 & & & \\
\text{subject to} & x_1 & +2x_2 & +3x_3 & & = & 3 \\
 & & -4x_2 & -9x_3 & & = & -5 \\
 & & & +3x_3 & +x_4 & = & 1 \\
 & x_1 & , x_2 & , x_3 & , x_4 & \ge & 0
\end{array}
$$

Finding an Initial BFS

How to find an initial BFS?

- One could test different bases B to see if $\boldsymbol{A}_B^{-1}\boldsymbol{b} \geq \boldsymbol{0}$.

$\rightsquigarrow$ This may take a long time!

- In fact, in terms of computational complexity (which we will define later), finding a BFS is as hard as finding the optimal solution!

We will discuss an initialization method — two-phase method.

Two-Phase Simplex Method

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbb{1}$ is an all-one vector):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{y}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

$\mathbb{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^m$

Without loss of generality, we can assume $\boldsymbol{b} \geq \boldsymbol{0}$ (otherwise, we pre-multiply that row by -1).

If $b_i < 0$, then we multiple i-th row of A by -1.

Two-Phase Simplex Method

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbb{1}$ is an all-one vector):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{y}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

Without loss of generality, we can assume $\boldsymbol{b} \geq \boldsymbol{0}$ (otherwise, we pre-multiply that row by -1).

- Trivial BFS for this auxiliary problem: $\boldsymbol{x} = \boldsymbol{0}$, $\boldsymbol{y} = \boldsymbol{b} \geq \boldsymbol{0}$.
- We can apply the simplex method to solve it.

Two-Phase Simplex Method

In the two-phase simplex method, we first solve an auxiliary problem ($\mathbb{1}$ is an all-one vector):

$$\begin{aligned} \text{minimize}_{\boldsymbol{x},\boldsymbol{y}} \quad & \mathbb{1}^\top \boldsymbol{y} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

Without loss of generality, we can assume $\boldsymbol{b} \geq \boldsymbol{0}$ (otherwise, we pre-multiply that row by -1).

- Trivial BFS for this auxiliary problem: $\boldsymbol{x} = \boldsymbol{0}$, $\boldsymbol{y} = \boldsymbol{b} \geq \boldsymbol{0}$.
- We can apply the simplex method to solve it.

Theorem: Feasibility

The original problem is feasible if and only if the optimal value of the auxiliary problem is 0. i.e., $y=0$

Proof of the Theorem. First, if the original problem is feasible with a feasible solution $\mathbf{x}_0$. Then $\mathbf{x} = \mathbf{x}_0$, $\mathbf{y} = 0$ is a feasible solution to the auxiliary problem with objective value 0.

Proof of the Theorem. First, if the original problem is feasible with a feasible solution $\mathbf{x}_0$. Then $\mathbf{x} = \mathbf{x}_0$, $\mathbf{y} = 0$ is a feasible solution to the auxiliary problem with objective value 0. Note that the optimal value of the auxiliary problem cannot be less than 0. Therefore the optimal value of the auxiliary problem must be 0. ("only if" part proved)

Proof of the Theorem. First, if the original problem is feasible with a feasible solution $\mathbf{x}_0$. Then $\mathbf{x} = \mathbf{x}_0$, $\mathbf{y} = 0$ is a feasible solution to the auxiliary problem with objective value 0. Note that the optimal value of the auxiliary problem cannot be less than 0. Therefore the optimal value of the auxiliary problem must be 0. ("only if" part proved)

Second, if the optimal value of the auxiliary problem is 0. Say $(\mathbf{x}^*, \mathbf{y}^*)$ is the optimal solution. Then it must be that $\mathbf{y}^* = 0$. Then $\mathbf{x}^*$ is a feasible solution to the original problem. □

Two-Phase Simplex Method

Hence, we can solve the auxiliary problem by the simplex method:

1. If the optimal value is not 0, then we can claim that the original problem is infeasible.

Two-Phase Simplex Method

Hence, we can solve the auxiliary problem by the simplex method:

1. If the optimal value is not 0, then we can claim that the original problem is infeasible.
2. If the optimal value is 0, then we can find an optimal BFS $(\boldsymbol{x}^*, \boldsymbol{y}^* = \mathbf{0})$ with basis B to the auxiliary problem.

Two-Phase Simplex Method

Hence, we can solve the auxiliary problem by the simplex method:

1. If the optimal value is not 0, then we can claim that the original problem is infeasible.
2. If the optimal value is 0, then we can find an optimal BFS $(\boldsymbol{x}^*, \boldsymbol{y}^* = \mathbf{0})$ with basis B to the auxiliary problem.
   - If $\boldsymbol{x}^*$ contains the full basis B, then x^* is a BFS for the orignal problem with basis B;

Two-Phase Simplex Method

Hence, we can solve the auxiliary problem by the simplex method:

1. If the optimal value is not 0, then we can claim that the original problem is infeasible.
2. If the optimal value is 0, then we can find an optimal BFS $(\boldsymbol{x}^*, \boldsymbol{y}^* = \mathbf{0})$ with basis B to the auxiliary problem.
   - If $\boldsymbol{x}^*$ contains the full basis B, then x^* is a BFS for the orignal problem with basis B;
   - If $\boldsymbol{x}^*$ does not contains the full basis B, (i.e., at least one basic index in $\boldsymbol{y}^* \rightarrow$ degenerate case), denote those basic indexes in $\boldsymbol{x}^*$ as $B(1), ..., B(k)$ with $k < m$. Then we can find a set $S = \{S(1), ..., S(m-k)\}$ disjoint from B such that columns of $A_{B \cup S}$ are linearly independent. Therefore, $B \cup S$ is a basis for $\boldsymbol{x}^*$ in the original problem.

Two-Phase Simplex Method

Hence, we can solve the auxiliary problem by the simplex method:

1. If the optimal value is not 0, then we can claim that the original problem is infeasible.
2. If the optimal value is 0, then we can find an optimal BFS $(\boldsymbol{x}^*, \boldsymbol{y}^* = \boldsymbol{0})$ with basis B to the auxiliary problem.
   - If $\boldsymbol{x}^*$ contains the full basis B, then x^* is a BFS for the orignal problem with basis B;
   - If $\boldsymbol{x}^*$ does not contains the full basis B, (i.e., at least one basic index in $\boldsymbol{y}^* \to$ degenerate case), denote those basic indexes in $\boldsymbol{x}^*$ as $B(1), ..., B(k)$ with $k < m$. Then we can find a set $S = \{S(1), ..., S(m-k)\}$ disjoint from B such that columns of $A_{B \cup S}$ are linearly independent. Therefore, $B \cup S$ is a basis for $\boldsymbol{x}^*$ in the original problem.

$\rightsquigarrow$ We can start from $\boldsymbol{x}^*$ and its basis to initialize the original simplex method.

Procedure of the Two-Phase Method

Phase I:

1. Construct the auxiliary problem such that $\boldsymbol{b} \geq \mathbf{0}$.
2. Solve the auxiliary problem using the simplex method.
   - If we reach an optimal solution with optimal value greater than 0, then the original problem is infeasible.
3. If the optimal value is 0 with an optimal solution $(\boldsymbol{x}^*, \boldsymbol{y}^*)$, then we enter phase II.

Phase II:

Solve the original problem starting from the BFS $\boldsymbol{x}^*$:

- If $(\boldsymbol{x}^*, \boldsymbol{y}^*)$ is degenerate for the auxiliary problem, then we need to supplement some indices to have a basis for the original problem.

The Big-M Method

There is another method that can be used to solve LP without a starting BFS. Consider the following auxiliary problem:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} + M \sum_{i=1}^{m} y_i \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

This problem has an initial BFS $\boldsymbol{x} = \boldsymbol{0}, \boldsymbol{y} = \boldsymbol{b} \geq \boldsymbol{0}$ (assuming $\boldsymbol{b} \geq \boldsymbol{0}$). $\rightsquigarrow$ We can use the simplex method to solve it.

The Big-M Method

There is another method that can be used to solve LP without a starting BFS. Consider the following auxiliary problem:

$$\begin{aligned} \text{minimize} \quad & \boldsymbol{c}^\top \boldsymbol{x} + M \sum_{i=1}^{m} y_i \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} + \boldsymbol{y} = \boldsymbol{b} \\ & \boldsymbol{x}, \boldsymbol{y} \geq \boldsymbol{0}. \end{aligned}$$

This problem has an initial BFS $\boldsymbol{x} = \boldsymbol{0}, \boldsymbol{y} = \boldsymbol{b} \geq \boldsymbol{0}$ (assuming $\boldsymbol{b} \geq \boldsymbol{0}$). $\rightsquigarrow$ We can use the simplex method to solve it.

In the simplex procedure, pretend that M is a very large value (larger than any specified number):

- If the original problem is feasible, then the optimal solution will have $\boldsymbol{y}^* = 0$ and does not involve basic indices in $\boldsymbol{y}$.
- The two-phase approach is more common. Expecially in the simplex tableau.

The Simplex Tableau

Simplex Tableau

New Goal:

- We want to have a simpler implementation of the simplex method

$\rightsquigarrow$ Avoid explicit matrix inversion in the calculations!

We are going to introduce the simplex tableau, which is a practical way to implement the simplex method.

- The simplex tableau maintains a table of numbers.
- It visualizes the procedures of the simplex algorithm and facilitates the computation.
- After learning the simplex tableau, one should be able to solve small-sized linear optimization problems by hand.

Simplex Tableau

- Let B the current basis with basis matrix $\boldsymbol{A}_B$ and basis objective coefficients $\boldsymbol{c}_B$.
- The simplex tableau is a table with the following structure:

reduced cost →

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$ ← negative of obj. function
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

↓

x_B

$c^T x = c_B^T x_B + c_N^T x_N$

$= c_B^T x_B$ ($c_N^T x_N = 0$)

$= c_B^T A_B^{-1} b$

Simplex Tableau

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

The lower part of the tableau can be viewed as a transformation of the constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ to $\boldsymbol{A}_B^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}_B^{-1}\boldsymbol{b}$:

- This is equivalent to the original constraint.

Simplex Tableau

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

The lower part of the tableau can be viewed as a transformation of the constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ to $\boldsymbol{A}_B^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}_B^{-1}\boldsymbol{b}$:

- This is equivalent to the original constraint.

Writing $\boldsymbol{A} = [\boldsymbol{A}_B, \boldsymbol{A}_N]$, we have

$$\boldsymbol{A}_B^{-1}\boldsymbol{A} = [\boldsymbol{I}, \boldsymbol{A}_B^{-1}\boldsymbol{A}_N].$$

- This block must contain an identity matrix.

Simplex Tableau

$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$	$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b}$
$\boldsymbol{A}_B^{-1} \boldsymbol{A}$	$\boldsymbol{A}_B^{-1} \boldsymbol{b}$

The lower part of the tableau can be viewed as a transformation of the constraint $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ to $\boldsymbol{A}_B^{-1}\boldsymbol{A}\boldsymbol{x} = \boldsymbol{A}_B^{-1}\boldsymbol{b}$:

- This is equivalent to the original constraint.

Writing $\boldsymbol{A} = [\boldsymbol{A}_B, \boldsymbol{A}_N]$, we have

$$\boldsymbol{A}_B^{-1}\boldsymbol{A} = [\boldsymbol{I}, \boldsymbol{A}_B^{-1}\boldsymbol{A}_N].$$

- This block must contain an identity matrix.

When the basis is B, the current basic feasible solution is

$$\boldsymbol{x} = [\boldsymbol{x}_B; \boldsymbol{x}_N] = [\boldsymbol{A}_B^{-1}\boldsymbol{b}; \boldsymbol{0}].$$

Therefore, the lower right corner yields the current BFS.

The term

$$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$$

a row vector

$(\bar{c}_1 \ \bar{c}_2 \ \cdots \ \bar{c}_n)$

is the reduced cost at this basis.

- Recall that the reduced costs for basic variables are zero. Therefore, this block contains zeros for the basic indices.

The term

$$\boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}$$

is the reduced cost at this basis.

- Recall that the reduced costs for basic variables are zero. Therefore, this block contains zeros for the basic indices.

Lastly, the term

$$-\boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{b} = -\boldsymbol{c}_B^\top \boldsymbol{x}_B$$

is the negative of the objective value at this basis.

Simplex Tableau

Thus, the simplex tableau should look like (after reordering the columns in $[B; N]$):

→ corresponds to B

$\mathbf{0}_m^\top$	$\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$	$-\boldsymbol{c}_B^\top \boldsymbol{x}_B$
$\boldsymbol{I}_m$	$\boldsymbol{A}_B^{-1} \boldsymbol{A}_N$	$\boldsymbol{x}_B$

Here, $\mathbf{0}_m \in \mathbb{R}^m$ is a vector of m zeros and $\boldsymbol{I}_m$ is the m-dimensional identity matrix.

Simplex Tableau

Thus, the simplex tableau should look like (after reordering the columns in $[B; N]$):

$\mathbf{0}_m^\top$	$\boldsymbol{c}_N^\top - \boldsymbol{c}_B^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N$	$-\boldsymbol{c}_B^\top \boldsymbol{x}_B$
$\boldsymbol{I}_m$	$\boldsymbol{A}_B^{-1} \boldsymbol{A}_N$	$\boldsymbol{x}_B$

Here, $\mathbf{0}_m \in \mathbb{R}^m$ is a vector of m zeros and $\boldsymbol{I}_m$ is the m-dimensional identity matrix.

This form of an LP is called the canonical form:

- The constraint matrix for the basic variables (not necessarily the first m columns) is an identity matrix.
- The objective function part for the basic variables is zero.

Canonical Form

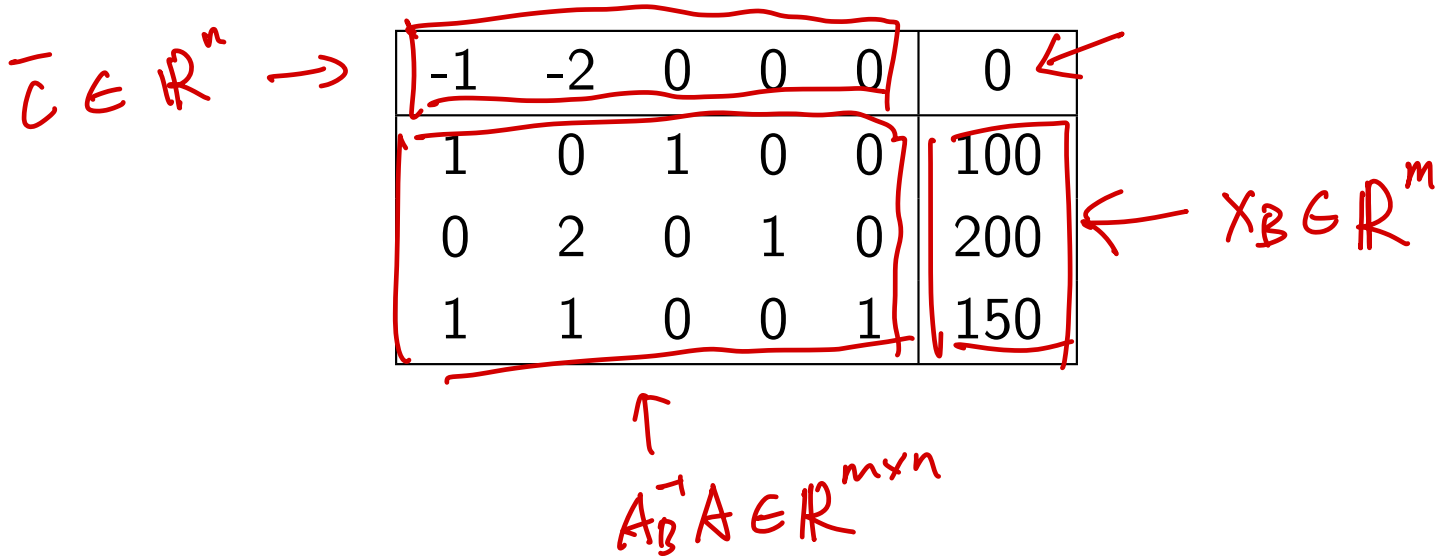
In the canonical form, it is easy to see whether the current BFS is optimal or not:

- ⇝ Consider the reduced costs in the top row of the tableau.
- If in a simplex tableau, the numbers in the top row are all nonnegative, then we have reached optimality!

Canonical Form

In the canonical form, it is easy to see whether the current BFS is optimal or not:

- ⇝ Consider the reduced costs in the top row of the tableau.
- If in a simplex tableau, the numbers in the top row are all nonnegative, then we have reached optimality!



In the canonical form, it is easy to see whether the current BFS is optimal or not:

- ⇝ Consider the reduced costs in the top row of the tableau.
- If in a simplex tableau, the numbers in the top row are all nonnegative, then we have reached optimality!

-1	-2	0	0	0	0
1	0	1	0	0	100
0	2	0	1	0	200
1	1	0	0	1	150

→ obj: 0

$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix} = \begin{pmatrix} 100 \\ 200 \\ 150 \\ 0 \\ 0 \end{pmatrix}$

- Are we in an optimal solution? ⇝ No.

↑

$\bar{c}_j < 0$ for some j.

Today's lecture

- Initialization of the simplex method
- Simplex tableau

{ Two-phase
  big M

Next lecture

- How to use Simplex tableau

Questions?